24. Predictive Maintenance Using Classification and Regression • Scenario: A manufacturing plant wants to predict machine failures. • Task: Use classification to detect failure types and regression to estimate time to-failure.

```
import numpy as np

from sklearn.model_selection import train_test_split

from sklearn.ensemble import RandomForestClassifier, RandomForestRegressor


# Synthetic dataset

# Features: [temperature, vibration, wear_level]

X = np.array([

[70,1200,0.2],

[85,1500,0.5],

[60,900,0.1],

[95,1800,0.7],

[75,1300,0.3],

[90,1700,0.6]

])


# Classification labels

# 0 = No failure, 1 = Minor failure, 2 = Major failure

y_class = np.array([0,1,0,2,0,2])


# Regression target

# Time to failure in hours

y_reg = np.array([120,40,150,20,100,30])


# Split dataset into training and testing sets

X_train, X_test, yc_train, yc_test, yr_train, yr_test = train_test_split(

   X, y_class, y_reg, test_size=0.3

)
```

```
# Train classification model to detect failure type
clf = RandomForestClassifier()
clf.fit(X_train, yc_train)

# Train regression model to estimate time to failure
reg = RandomForestRegressor()
reg.fit(X_train, yr_train)

# New machine data for prediction
sample = np.array([[88,1600,0.55]])

# Predict failure type
print("Predicted Failure Type:", clf.predict(sample))

# Predict remaining time to failure
print("Estimated Time to Failure:", reg.predict(sample))
```